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CST476 MOBILE COMPUTING - Module II

CST476 MOBILE COMPUTING

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Spread Spectrum in Mobile Computing

Spread spectrum is a technique used for wireless communications in telecommunication and radio communication. The frequency of the transmitted signal is deliberately varied and generates a much greater bandwidth than the signal would have if its frequency were not varied.

Spread Spectrum is a technique in which the transmitted signals of specific frequencies are varied slightly to obtain greater bandwidth as compared to initial bandwidth.

Spread spectrum technology is widely used in radio signals transmission because it can easily reduce noise and other signal issues.

The spread spectrum model is used to overcome with this conventional communication model. Here, the transmitted signal frequency is deliberately varied over a comparatively large segment of the electromagnetic radiation spectrum. This variation is done according to a specific but complicated mathematical function. If the receiver wants to intercept the signal, it must be tuned to frequencies that vary precisely according to this function.

Reasons to use Spread Spectrum

1. Spread spectrum signals are distributed over a wide range of frequencies and then collected and received back to the receiver.
2. Initially, the spread spectrum was adopted in military applications because of its resistance to jamming and difficulty intercepting.
3. Now, this is also used in commercial wireless communication. It is most preferred because of its useful bandwidth utilization ability.

Characteristics of the Spread Spectrum are:

- Higher channel capacity.
- Ability to resist multi-path propagation.
- They cannot easily intercept any unauthorized person.
- They are resistant to jamming.
- The spread spectrum offers multiple access capabilities.

Types of Spread Spectrum

Spread Spectrum can be categorized into two types:

1. Frequency Hopping Spread Spectrum (FHSS)
2. Direct Sequence Spread Spectrum (DSSS)



Frequency Hopping Spread Spectrum (FHSS)

In Frequency Hopping Spread Spectrum (FHSS), different carrier frequencies are modulated by the source signal. At one moment signal modulates one carrier frequency and at the subsequent moments, it modulates other carrier frequencies.

- The Frequency Hopping Spread Spectrum or FHSS allows us to utilize bandwidth properly and maximum. In this technique, the whole available bandwidth is divided into many channels and spread between channels, arranged continuously.
- The frequency slots are selected randomly, and frequency signals are transmitted according to their occupancy.
- The transmitters and receivers keep on hopping on channels available for a particular amount of time in milliseconds.
- It is possible to implement the frequency division multiplexing and time-division multiplexing simultaneously in FHSS.

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The Frequency Hopping Spread Spectrum or FHSS can also be classified into two types:

1. **Slow Hopping:** In slow hopping, multiple bits are transmitted on a specific frequency or same frequency.
2. **Fast Hopping:** In fast hopping, individual bits are split and then transmitted on different frequencies.

Advantages of Frequency Hopping Spread Spectrum (FHSS)

1. The biggest advantage of Frequency Hopping Spread Spectrum or FHSS is its high efficiency.
2. The Frequency Hopping Spread Spectrum or FHSS signals are highly resistant to narrowband interference because the signal hops to a different frequency band.
3. It requires a shorter time for acquisition.
4. It is highly secure.
5. It provides a very large bandwidth.
6. It can be simply implemented as compared to other - DSSS.

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Disadvantages of Frequency Hopping Spread Spectrum (FHSS)

1. FHSS is less Robust
2. FHSS needs complex frequency synthesizers.

3. FHSS supports a lower data rate of 3 Mbps as compared to the 11 Mbps data rate supported by DSSS.
4. It is not very useful for range and range rate measurements.
5. It supports the lower coverage range

Applications of Frequency Hopping Spread Spectrum (FHSS)

1. The Frequency Hopping Spread Spectrum or FHSS is used in wireless local area networks (WLAN) standard for Wi-Fi.
2. FHSS is also used in the wireless personal area networks (WPAN) standard for Bluetooth.

Direct Sequence Spread Spectrum (DSSS)

The Direct Sequence Spread Spectrum (DSSS) is a spread-spectrum modulation technique primarily used to reduce overall signal interference in telecommunication. The Direct Sequence Spread Spectrum modulation makes the transmitted signal wider in bandwidth than the information bandwidth.

- In DSSS, the message bits are modulated by a bit sequencing process known as a spreading sequence.
- This spreading-sequence bit is known as a chip.
- It has a much shorter duration (larger bandwidth) than the original message bits.

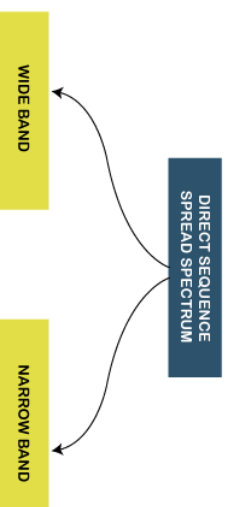
Features of Direct Sequence Spread Spectrum (DSSS).

1. In Direct Sequence Spread Spectrum or DSSS technique, the data that needs to be transmitted is split into smaller blocks. After that, each data block is attached with a high data rate bit sequence and is transmitted from the sender end to the receiver end.
2. Data blocks are recombined again to generate the original data at the receiver's end, which was sent by the sender, with the help of the data rate bit sequence.

3. If somehow data is lost, then data blocks can also be recovered with those data rate bits.
4. The main advantage of splitting the data into smaller blocks is that it reduces the noise and unintentional inference.

The Direct Sequence Spread Spectrum or DSSS can also be classified into two types:

1. Wide Band Spread Spectrum
2. Narrow Band Spread Spectrum



Advantages of Direct Sequence Spread Spectrum (DSSS)

1. Direct Sequence Spread Spectrum or DSSS is less reluctant to noise; that's why the DSSS system's performance in the presence of noise is better than the FHSS system.
2. In Direct Sequence Spread Spectrum or DSSS, signals are challenging to detect.
3. It provides the best discrimination against multipath signals.
4. In Direct Sequence Spread Spectrum, there are very few chances of jamming because it avoids intentional interference such as jamming effectively.

Disadvantages of Direct Sequence Spread Spectrum (DSSS)

1. The Direct Sequence Spread Spectrum or DSSS system takes large acquisition time; that's why its performance is slow.
2. It requires wide-band channels with small phase distortion.
3. In DSSS, the pseudo-noise generator generates a sequence at high rates.

Applications of Direct Sequence Spread Spectrum (DSSS)

1. Direct Sequence Spread Spectrum or DSSS is used in LAN technology.
2. Direct Sequence Spread Spectrum or DSSS is also used in Satellite communication technology.
3. DSSS is used in the military and many other commercial applications.
4. It is used in the low probability of the intercept signal.
5. It supports Code division multiple access.

Media Access Control (MAC)

The Media Access Control (MAC) data communication protocol sub-layer, also known as the Medium Access Control, is a sublayer of the Data Link Layer specified in the seven-layer OSI model (layer 2)

The hardware that implements the MAC is referred to as a Medium Access Controller.

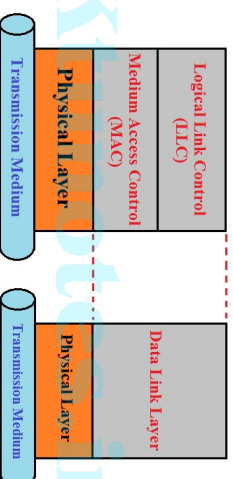


Figure 1: LLC and MAC sub-layer

The MAC sub-layer acts as an interface between the Logical Link Control (LLC) sublayer and the network's physical layer. The MAC layer emulates a full-duplex logical communication channel in a multi-point network. This channel may provide unicast, multicast or broadcast communication service.

One of the most commonly used MAC schemes for wired networks is carrier sense multiple access with collision detection (CSMA/CD). In this scheme, a sender senses the medium (a wire or coaxial cable) to see if it is free. If the medium is busy, the sender waits until it is free. If the medium is free, the sender starts transmitting data and continues to listen into the medium. If the sender detects a collision while sending, it stops at once and sends a jamming signal. But this scheme does not work well with wireless networks.

The problems are:

- Signal strength decreases proportional to the square of the distance
- The sender would apply CS and CD, but the collisions happen at the receiver
- It might be a case that a sender cannot listen the collision, i.e., CD does not work
- Furthermore, CS might not work, if for e.g., a terminal is hidden.

The **near/far** effect is a severe problem of wireless networks using CDMA. All signals should arrive at the receiver with more or less the same strength for which Precise power control is to be implemented.

Space Division Multiple Access (SDMA)

Space Division Multiple Access (SDMA) is used for allocating a separated space to users in wireless networks. A typical application involves assigning an optimal base station to a mobile phone user. The mobile phone may receive several base stations with different quality. A MAC algorithm could now decide which base station is best, taking into account which frequencies (FDM), time slots (TDM) or code (CDM) are still available.

The basis for the SDMA algorithm is formed by cells and sectorized antennas which constitute the infrastructure implementing space division multiplexing (SDM).

SDM has the unique advantage of not requiring any multiplexing equipment.

It is usually combined with other multiplexing techniques to better utilize the individual physical channels.

Frequency division multiplexing (FDM)

Frequency division multiplexing (FDM) describes schemes to subdivide the frequency dimension into several non-overlapping frequency bands.

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non-overlapping frequency bands. Frequency Division Multiple Access is a method employed to permit several users to transmit simultaneously on one satellite transponder by assigning a specific frequency within the channel to each user. Each conversation gets its own, unique, radio channel. The channels are relatively narrow, usually 30 KHz or less and are defined as either transmit or receive channels. A full duplex conversation requires a transmit & receive channel pair. FDM is often used for simultaneous access to the medium by base station and mobile station in cellular networks establishing a duplex channel.

A scheme called frequency division duplexing (FDD) in which the two directions, mobile station to base station and vice versa are now separated using different frequencies.

The two frequencies are also known as **uplink**, i.e., from mobile station to base station or from ground control to satellite, and as **downlink**, i.e., from base station to mobile station or from satellite to ground control. The basic frequency allocation scheme for GSM is fixed and regulated by national authorities. All uplinks use the band between 890.2 and 915 MHz, all downlinks use 935.2 to 960 MHz.

This scheme also has disadvantages. While radio stations broadcast 24 hours a day, mobile communication typically takes place for only a few minutes at a time. Assigning a separate frequency for each possible

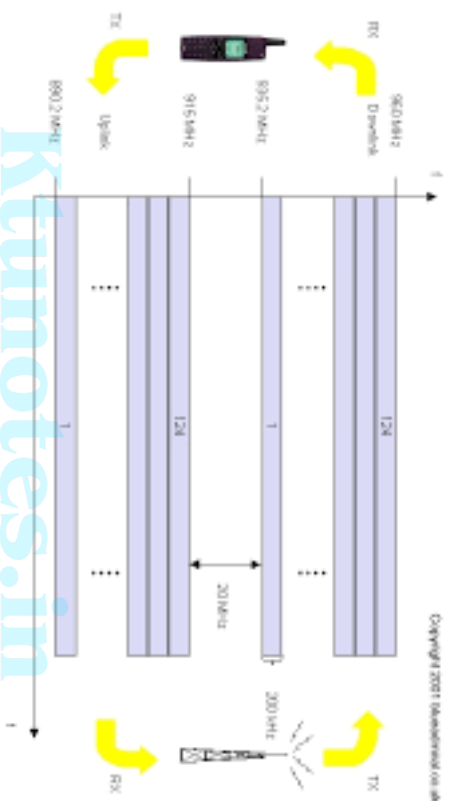


Figure 2: FDM for multiple access and duplex

communication scenario would be a tremendous waste of (scarce) frequency resources. Additionally, the fixed assignment of a frequency to a sender makes the scheme very inflexible and limits the number of senders.

Time Division Multiplexing (TDM)

A more flexible multiplexing scheme for typical mobile communications is time division multiplexing (TDM). Compared to FDMA, time division multiple access (TDMA) offers a much more flexible scheme, which comprises all technologies that allocate certain time slots for communication. Now synchronization between sender and receiver has to be achieved in the time domain. Again this can be done by using a fixed pattern similar to FDMA techniques, i.e., allocating a certain time slot for a channel, or by using a dynamic allocation scheme.

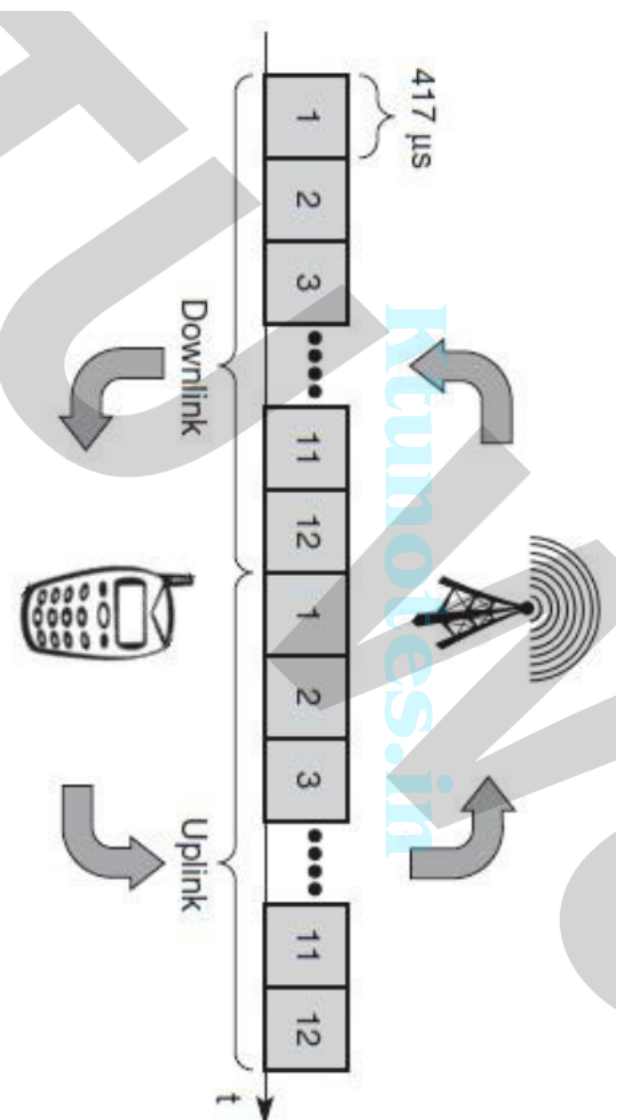
Note:

Listening to different frequencies at the same time is quite difficult, but listening to many channels separated in time at the same frequency is simple. Fixed schemes do not need identification, but are not as flexible considering varying bandwidth requirements.

Fixed TDM

The simplest algorithm for using TDM is allocating time slots for channels in a fixed pattern. This results in a fixed bandwidth and is the typical solution for wireless phone systems. MAC is quite simple, as the only

crucial factor is accessing the reserved time slot at the right moment. If this synchronization is assured, each mobile station knows its turn and no interference will happen. The fixed pattern can be assigned by the base station, where competition between different mobile station that want to access the medium is solved.



The above figure shows how these fixed TDM patterns are used to implement multiple access and a duplex channel between a base station and mobile station. Assigning different slots for **uplink** and **downlink** using the same frequency is called time division duplex (TDD). As shown in the figure, the base station uses one out of 12 slots for the downlink, whereas the mobile station uses one out of 12 different slots for the uplink. Uplink and downlink are separated in time. Up to 12 different mobile stations can use the same frequency without interference using this scheme. Each connection is allotted its own up- and downlink pair. This general scheme still wastes a lot of bandwidth. It is too static, too inflexible for data communication. In this case, connection less, demand-oriented TDMA schemes can be used

Code division multiple access (CDMA)

Code division multiple access systems apply codes with certain characteristics to the transmission to separate different users in code space and to enable access to a shared medium without interference.

All terminals send on the same frequency probably at the same time and can use the whole bandwidth of the transmission channel. Each sender has a unique random number, the sender XORs the signal with this random number.

The receiver can **tune** into this signal if it knows the pseudo random number, tuning is done via a correlation function.

Disadvantages:

- ☐ Higher complexity of a receiver (receiver cannot just listen into the medium and start receiving if there is a signal)
- ☐ All signals should have the same strength at the receiver.

Advantages:

- ☐ All terminals can use the same frequency, no planning needed.
- ☐ Huge code space (e.g. 2^{32}) compared to frequency space
- ☐ Interference (e.g. white noise) is not coded
- ☐ Forward error correction and encryption can be easily integrated

Soft handover or **soft handoff** refers to a feature used by the CDMA and WCDMA standards, where a cell phone is simultaneously connected to two or more cells (or cell sectors) during a call. If the sectors are from the same physical cell site (a sectorised site), it is referred to as softer handoff. This technique is a

form of mobile-assisted handover, for IS-95/CDMA2000 CDMA cell phones continuously make power measurements of a list of neighboring cell sites, and determine whether or not to request or end soft handover with the cell sectors on the list.

Soft handoff is different from the traditional hard-handoff process. With hard handoff, a definite decision is made on whether to hand off or not. The handoff is initiated and executed without the user attempting to have simultaneous traffic channel communications with the two base stations. With soft handoff, a conditional decision is made on whether to hand off. Depending on the changes in pilot signal strength from the two or more base stations involved, a hard decision will eventually be made to communicate with only one. This normally happens after it is evident that the signal from one base station is considerably stronger than those from the others. In the interim period, the user has simultaneous traffic channel communication with all candidate base stations. It is desirable to implement soft handoff in power-controlled CDMA systems because implementing hard handoff is potentially difficult in such systems.

Satellite Systems

- A satellite is an artificial object which is placed intentionally into an orbit of any natural satellite. Satellites are used for many purposes i.e. weather forecasting, digital transmission, scientific research and development etc.
- In a communication context, a satellite is a specialized wireless transmitter/receiver that is launched by a rocket and placed in orbit around the earth.
- A satellite can be natural, like the moon, or artificial (human made). So we can say that a satellite is an object that moves in a curved path around a planet.
- Satellite can travel around planets or around stars such as our sun. All the planets are satellites around the sun.
- Satellites which are far away from the surface of the earth can cover a wide area on the surface of the earth.
- The moon is earth's original, natural satellite, and there are many man-made (artificial) satellites for example: SPUTNIK, ECHO, and SYNCOM.

Applications of Satellite Systems

Traditionally, satellites have been used in the following areas:

1. **Weather forecasting:** Without the help of satellites, we cannot predict weather. Satellites have the biggest contribution to making predictions about weather changes by studying the several global scenarios.
Weather forecasting is done through fitting special instruments and powerful cameras in the satellites which monitor various climate factors such as air pressure, air temperature and humidity etc.
The satellites made for weather forecasting are called weather satellites
2. **Radio and TV broadcast satellites:** Hundreds of TV and radio programs are available via satellites. Today's satellite dishes have diameters of 30-40 cm in central Europe, and the diameters in northern countries are slightly larger.
3. **Military satellites:** One of the oldest applications of satellites was their use for carrying out espionage. Most of the communication links are managed via satellite because they are much safer from attack by enemies.

4. **Satellites for Navigation:** Even though it was only used for military purposes in the beginning, the GPS (Global Positioning System) is nowadays well known and available for everyone.

At our navigation system, Google maps etc allows for precise localization worldwide, and with some additional techniques, the precision is in the range of some meters.

Almost all aircraft and ships rely on GPS as an addition to traditional navigation systems. Many cars and trucks come with installed GPS receivers. This system is also used for fleet management of trucks or for vehicle localization in case of theft.

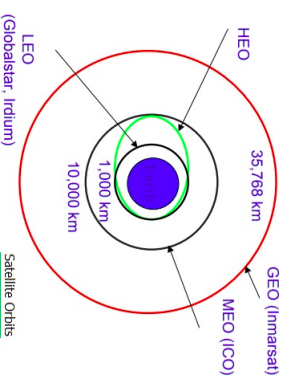
5. **Satellites for Communications:** Global mobile communication - The latest trend of satellites supports the global mobile data communication. Due to the high latency, geostationary satellites are not ideal for this task; therefore satellites using lower orbits are needed. The basic purpose of satellites for mobile communication is not to replace the existing mobile phone networks, but to extend the area of coverage.

Types of Satellite Systems

Satellites have been put in space for various purposes and their placement in space and orbiting shapes have been determined as per their specific requirements.

Four different types of satellites orbits have been identified. - These are:

- GEO (Geostationary Earth Orbit) at about 36,000km above the earth's surface.
- LEO (Low Earth Orbit) at about 500-1500km above the earth's surface.
- MEO (Medium Earth Orbit) at about 6000-20,000 km above the earth's surface.
- HEO (Highly Elliptical Orbit)



1. GEO (Geostationary Earth Orbit)

- ☐ If a satellite should appear in fixed in the sky, it requires a period of 24 hours. The distance is 35,786 km. above the earth and the orbit must have an inclination of 0 degree.
- ☐ Geostationary satellites have a distance of almost 36,000 km to the earth. Examples are almost all TV and radio broadcast satellites, any weather satellites and satellites operating as backbones for the telephone network.
- ☐ Objects in GEO moves around the earth at the same speed as the earth rotates. This means geostationary satellites remain in the same position relative to the surface of earth.

Advantages of GEO satellite

- ☐ Three Geostationary satellites are enough for a complete coverage of almost any spot on earth.
- ☐ Receivers and senders can use fixed antenna positions, no adjusting is needed.
- ☐ GEOs are ideal for TV and radio broadcasting.
- ☐ Lifetime expectations for GEOs are rather high, at about 15 years.
- ☐ Geostationary satellites have a 24 hour view of a particular area.

- ☐ GEOs typically do not need handover due to the large footprints.
- ☐ GEOs don't exhibit any Doppler shift because the relative movement is zero.

Disadvantages of GEO satellite

- ☐ Northern or southern regions of the earth have more problems receiving these satellites due to the low elevation above latitude of 60 degree, i.e. larger antennas are needed in this case.
- ☐ Shading of the signals in cities due to high buildings and the low elevation further away from the equator limits transmission quality.
- ☐ The transmit power needed is relatively high (about 10 W) which causes problems for battery powered devices.
- ☐ These satellites can't be used for small mobile phones.
- ☐ The biggest problem for voice and also data communication is high latency of over 0.25s one way-retransmission schemes which are known from fixed networks fail.
- ☐ Transferring a GEO into orbit is very expensive.

2. LEO (Low Earth Orbit)

- As LEOs circulate on a lower orbit, it is obvious that they exhibit a much shorter period (the typical duration of LEO periods are 95 to 120 minutes). Additionally, LEO systems try to ensure a high elevation for every spot on earth to provide a high quality communication link.
- Each LEO satellite will only be visible from the earth for about ten minutes.
- A further classification of LEOs into little LEOs with low bandwidth services (some 100 bit/s), big LEOs (some 1,000 bit/s) and broadband LEOs with plans reaching into the Mbits/s range can be found in Comparetto (1997).
- LEO satellites are much closer to earth than GEO satellites, ranging from 500 to 1,500 km above the surface.
- LEO satellites do not stay in fixed position relative to the surface, and are only visible for 15 to 20 minutes each pass.

Advantages of LEO satellite

- ☐ Using advanced compression schemes, transmission rates of about 2,400 bit/s can be enough for voice communication.
- ☐ LEOs even provide this bandwidth for mobile terminals with omni-directional antennas using low transmit power in the range of 1W.
- ☐ A LEO satellite smaller area of coverage is less of a waste of bandwidth.
- ☐ Using advanced compression schemes, transmission rates of about 2,400 bit/s can be enough for voice communication.
- ☐ A LEO satellite's proximity to earth compared to a Geostationary satellite gives it a better signal strength and less of a time delay, which makes it better for point to point communication.
- ☐ Smaller footprints of LEOs allow for better frequency reuse, similar to the concepts used for cellular networks.

Disadvantages of LEO satellite

- ☐ The biggest problem of the LEO concept is the need for many satellites if global coverage is to be reached.
- ☐ The high number of satellites combined with the fast movement's results in a high complexity of the whole satellite system.
- ☐ The short time of visibility with a high elevation requires additional mechanism for connection handover between different satellites.
- ☐ One general problem of LEO is the short lifetime of about five to eight years due to atmospheric drag and radiation from the inner Van Allen belt.
- ☐ The low latency via a single LEO is only half of the story.
- ☐ Other factors are the need for routing of data packets from satellite to satellite (or several times from base stations to satellites and back) if a user wants to communicate around the world.
- ☐ A GEO typically does not need this type of routing, as senders and receivers are most likely in the same footprints.

3. MEO (Medium Earth Orbit)

- ☐ A MEO satellite situates in orbit somewhere between 6,000 km to 20,000 km above the earth's surface.
- ☐ MEO satellites are similar to LEO satellites in the context of functionality.
- ☐ MEO satellites are similar to LEO satellite in functionality.
- ☐ Medium earth orbit satellites are visible for much longer periods of time than LEO satellites usually between 2 to 8 hours.
- ☐ MEO satellites have a larger coverage area than Low Earth Orbit satellites.
- ☐ MEOs can be positioned somewhere between LEOs and GEOs, both in terms of their orbit and due to their advantages and disadvantages.

Advantages of MEO

- ☐ Using orbits around 10,000km, the system only requires a dozen satellites which is more than a GEO system, but much less than a LEO system.
- ☐ These satellites move more slowly relative to the earth's rotation allowing a simpler system design (satellite periods are about six hours).

- Depending on the inclination, a MEO can cover larger populations, so requiring fewer handovers.
- A MEO satellite's longer duration of visibility and wider footprint means fewer satellites are needed in a MEO network than a LEO network.

Disadvantages of MEO

- Again due to the larger distance to the earth, delay increases to about 70-80 ms.
- The satellites need higher transmit power and special antennas for smaller footprints.
- A MEO satellite's distance gives it a longer time delay and weaker signal than LEO satellite.

4. HEO (High Earth Orbit)

- The High Earth orbit satellite is the only non-circular orbit of the four types.
- HEO satellite operates with an elliptical orbit, with a maximum altitude (apogee) similar to GEO, and a minimum altitude (perigee) similar to the LEO.
- The HEO satellites used for the special applications where coverage of high latitude locations is required.

Summary of Satellite Systems

- Satellites have been launched for various applications and are placed at different altitudes. Moreover their weights are also dissimilar.
- Provides connections between the Earth-based base station and the satellites.
- The GEO satellites, which are at an altitude of 35,768 km, orbit in the equatorial plane with zero degree inclination and complete exactly one rotation in a day.
- The antennas are at fixed positions, and an uplink band (reverse band) of 1634.5 to 1660.5 MHz and a downlink band (forward band) in the range of 1530 to 1559 MHz, are employed.
- Ku band frequencies (11 and 13 GHz) are employed for connection between the base station (earth station) and the satellites.
- A satellite typically has a large footprint, which can be up to 34% of the earth's surface covered, and therefore it is difficult to reuse frequencies.
- The global coverage of small mobile phones and data transmission typically cause high latency in the range of about 275 ms.
- LEO satellite are divided into little and big satellites.

- Little LEOs are smaller in size and are in the frequency range of 148 to 150.
- They cause alphanumeric displays at low bit rates (of the order of 1 kb/s) for two-way message and positioning information.
- Big LEO satellites have adequate power and bandwidth to provide various global mobile services (i.e. data transmission, paging, facsimile, and position location) along with good quality voice services for mobile systems such as handheld devices and vehicular transceivers.
- Big LEO transmits in the frequency range of 1610 to 1626.5 MHz (uplink) and 2483.5 to 2500 MHz (downlink) and orbit at about 500 to 1500 km above the earth surface.
- The latency is around 5 to 10 ms, and the satellite is visible for about 10 to 40 ms.
- Several satellites are needed to ensure global coverage. The same frequency spectrum is also used by MEO and GEO.
- In MEO systems, the slow-moving satellites orbit at a height of about 5,000 to 12,000 km above the earth and have a latency of about 70 to 80 ms.
- Specialized antennas are used to provide smaller footprints and higher transmitting power.

GSM (Global System for Mobile Communication)

Global System for Mobile Communication (GSM) is a digital cellular technology used for transmitting mobile voice and data services.

- ☐ The concept of GSM emerged from a cell-based mobile radio system at Bell Laboratories in the early 1970s.
- ☐ GSM is the name of a standardization group established in 1982 to create a common European mobile telephone standard.
- ☐ GSM is the most widely accepted standard in telecommunications and it is implemented globally.
- ☐ GSM owns a market share of more than 70 percent of the world's digital cellular subscribers.
- ☐ GSM makes use of narrowband Time Division Multiple Access (TDMA) technique for transmitting signals.
- ☐ GSM digitizes and compresses data, then sends it down through a channel with two other streams of user data, each in its own timeslot.

Features of GSM

- ☐ Improved spectrum efficiency
- ☐ International roaming
- ☐ Low-cost mobile sets and base stations (BSs)
- ☐ High-quality speech (voice)
- ☐ Compatibility with Integrated Services Digital Network (ISDN) and other telephone company services
- ☐ Support for new services
- ☐ Low powered handheld devices.
- ☐ International ISDN compatibility.

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GSM is divided into further 3 subsystems

1. BSS (Base Station Subsystem)
2. NSS (Network and Switching Subsystem)
3. OSS (Operating Subsystem)

BSS (Base Station Subsystem) handles traffic and signaling between a mobile phone and the network switching subsystem. BSS having two components BTS and BSC.

NSS (Network and Switching Subsystem) is the core network of GSM. That carried out call and mobility management functions for mobile phone present in network. NSS have different components like VLR, HLR and EIR.

OSS (Operating Subsystem) is a functional entity which the network operator monitor and control the system. OMC is the part of OSS. Purpose of OSS is to offer the customer cost-effective support for all GSM related maintenance services.

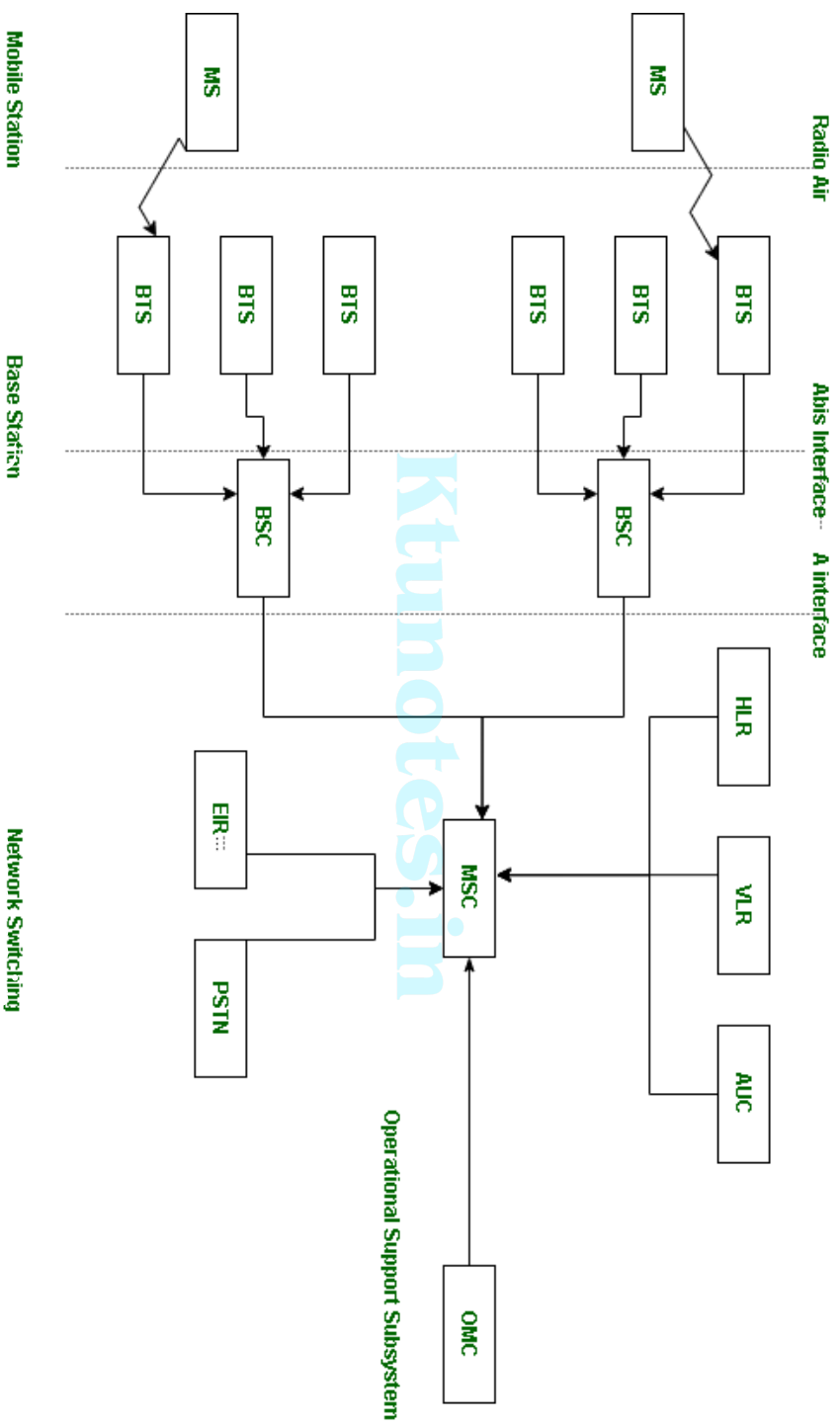
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GSM - Architecture

A GSM network comprises of many functional units. These functions and interfaces are explained in this chapter. The GSM network can be broadly divided into:

1. The Mobile Station (MS)
2. The Base Station Subsystem (BSS)
3. The Network Switching Subsystem (NSS)
4. The Operation Support Subsystem (OSS)

1. MS : MS stands for Mobile System. MS comprises user equipment and software needed for communication with a mobile network. Mobile Station (MS) = Mobile Equipment(ME) + Subscriber Identity Module (SIM). Now, these mobile stations are connected to tower and that tower connected with BTS through TRX. TRX is a transceiver which comprises transmitter and receiver. Transceiver has two performance of sending and receiving.
2. BTS : BTS stands for Base Transceiver Station which facilitates wireless communication between user equipment and a network. Every tower has BTS.
3. BSC : BSC stands for Base Station Controller. BSC has multiple BTS. You can consider the BSC as a local exchange of your area which has multiple towers and multiple towers have BTS.
4. MSC : MSC stands for Mobile Switching Center. MSC is associated with communication switching functions such as call setup, call release and routing. Call tracing, call forwarding all functions are performed at the MSC level. MSC is having further components like VLR, HLR, AUC, EIR and PSTN.
 - VLR : VLR stands for Visitor Location Register. VLR is a database which contains the exact location of all mobile subscribers currently present in the service area of MSC. If you are going from one state to another state then your entry is marked into the database of VLR.



□ HLR : HLR stands for Home Location Register. HLR is a database containing pertinent data regarding subscribers authorized to use a GSM network.. If you purchase SIM card from in the HLR. HLR is like a home which contains all data like your ID proof, which plan you are taking, which caller tune you are using etc.

□ OMC : OMC stands for Operation Maintenance Center. OMC monitor and maintain the performance of each MS, BSC and MSC within a GSM system.

□ AUC : AUC stands for Authentication Center. AUC authenticates the mobile subscriber that wants to connect in the network.

□ EIR : EIR stands for Equipment Identity Register. EIR is a database that keeps the record of all allowed or banned in the network. If you are banned in the network then you can't enter the network, and you can't make the calls.

□ PSTN : PSTN stands for Public Switched Telephone Network. PSTN connects with MSC. PSTN originally a network of fixed line analog telephone systems. Now almost entirely digital in its core network and includes mobile and other networks as well as fixed telephones. The earlier landline phones which places at our home is nothing but PSTN.

